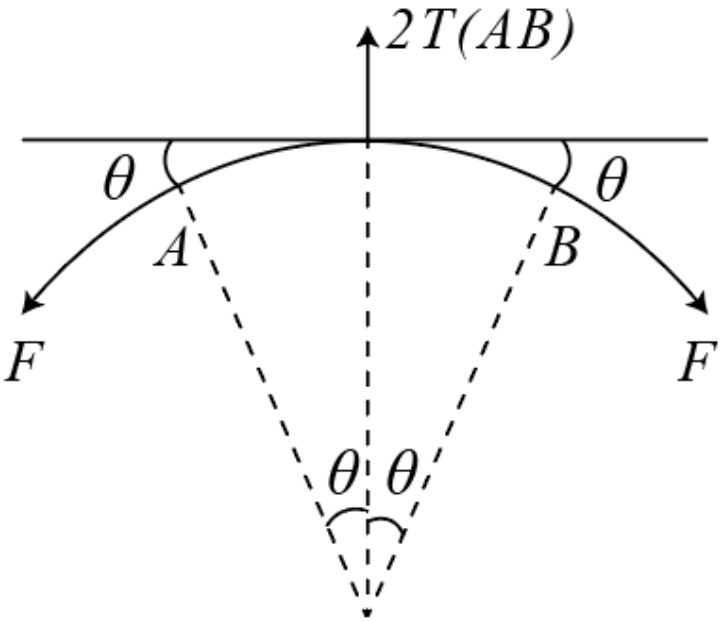
QPT-5 Solution

1. **Ans: (b)** $2RT$

Solution: $F =$ Tension in the thread
$2F\sin\theta = T(AB)$
$2F\theta = 2T(2R\theta)$



2. **Ans: (a)** 2rads^{-1}

Solution: In SHM,
Maximum velocity $v_{\max} = \omega A$
Maximum acceleration $a_{\max} = A\omega^2$

$$\Rightarrow \omega = \frac{A\omega^2}{A\omega} = \frac{a_{\max}}{v_{\max}}$$

$v_{\max} = 0.5\ \text{m s}^{-1}$

$a_{\max} = 1.0\ \text{m s}^{-1}$

$$\omega = \frac{1}{0.5} = \frac{10}{5} = 2\text{rads}^{-1}$$

3. **Ans: (b)** Wire Y is more ductile than wire X.

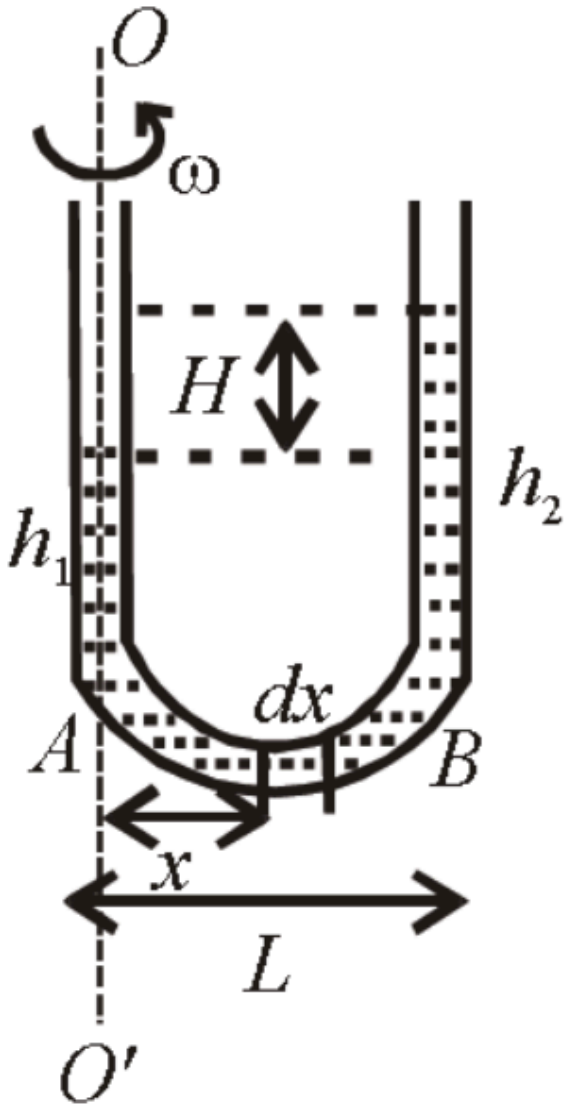
Solution: A metal is more ductile (and thus less brittle) when point of ultimate strength (i.e., C) and breaking point (i.e., D) are far apart.

So, option (b) is correct. Clearly, Young's modulus $\left(=\frac{\text{stress}}{\text{strain}}\right)$ of wire Y is more than wire X

4. **Ans: (a)** water levels in both sections A and B go up

Solution: Due to rotation about the axis OO', a force is acting on the liquid column AB outwards, pushing up the liquid column on the right. This centrifugal force acting on the small element dx of area of cross-section dx, distant x from the axis is
$dm\omega^2 x = Adx\rho\omega^2 x$
The total force due to column of length $L(=AB)$ is
$\int_0^L A\rho\omega^2 \cdot x dx = A\rho\omega^2\ \frac{L^2}{2}$

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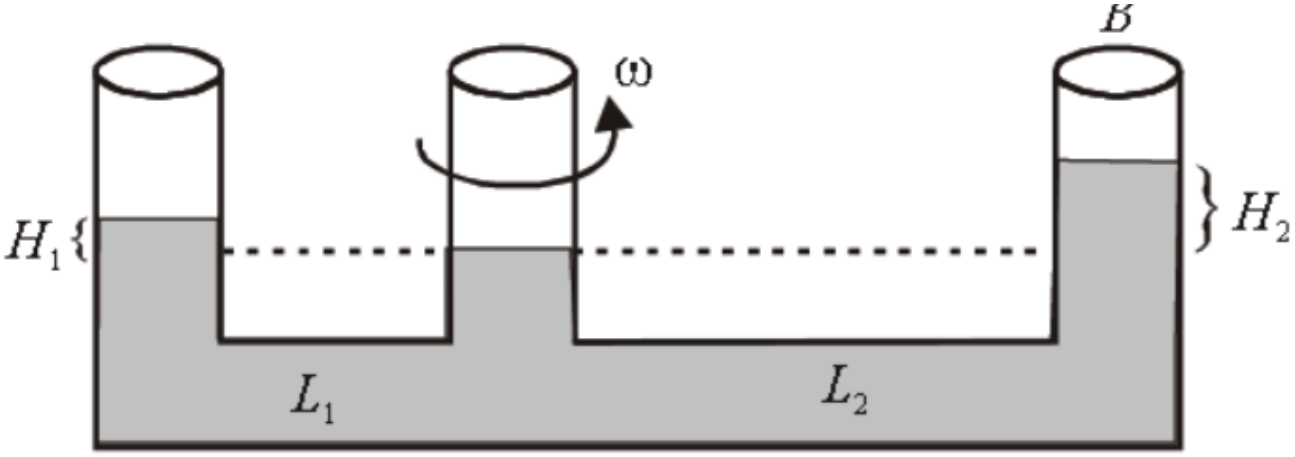


Pressure at B = atmospheric pressure +
$h_1 \rho g + \frac{A\rho\omega^2}{A} \frac{L^2}{2}$
Pressure at B due to liquid column on the right hand side = atmospheric pressure $+h_2 \rho g$
Net pressure due to the left hand side = pressure at B due to liquid column

$$\text{atm. pressure } + h_1 \rho g + \rho\omega^2 \frac{L^2}{2} = \text{ atm. pressure } + h_2 \rho g$$

$$\therefore h_2 - h_1 = H_0 = \frac{\omega^2 L^2}{2g} .$$

Here one has



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$H_2 = \frac{\omega^2 L_2^2}{2g}, H_1 = \frac{\omega^2 L_1^2}{2g}$
As ω is the same, and $L_2 > L_1, H_2 > H_1$. But both will go up.

5. **Ans: (d)** $40\ \text{cms}^{-1}$

Solution:

Terminal velocity is proportional to square of radius of the sphere.

$v \propto r^2$
$\frac{v_1}{v_2} = \frac{r_1^2}{r_2^2}$

and using volume conservation,

$8 \times \frac{4}{3}\pi r_1^3 = \frac{4}{3}\pi r_2^3$
$\Rightarrow r_2 = 2r_1$

therefore, $\text{v}_2 = 4\text{v}_1 = 40\ \text{cms}^{-1}$

6. **Ans: (c)** 5

Solution: Given, the position and velocity of the particle executing SHM.
$y = 3\ \text{cm}$
$v = 8\ \text{cm s}^{-1}$
Angular frequency, $\omega = 2\text{rad s}^{-1}$
The velocity,
$v = \omega\sqrt{a^2 - y^2}$
$8 = 2\sqrt{a^2 - (3)^2}$
$4 = \sqrt{a^2 - (3)^2}$
$16 = a^2 - 9$
$a^2 = 25$
$a = 5\ \text{cm}$

7. **Ans: (d)** Assertion is false but, reason is true.

Solution:

Given,

Breaking stress is equal for both the wires.

$\text{Breaking stress} = \frac{\text{Breaking load}}{\text{Cross}-\text{sectional area}}$.

Thus, the breaking load depends on the cross-sectional area of the wire if the breaking stress is same.

Therefore, breaking load of two wires can be different and the lower wire may have much lesser breaking load as compared with upper wire.

8. **Ans: (d)** All cases

Solution: This happens in accordance with equation of continuity and this equation was derived on the principle of conservation of mass and it is true in every case, either tube remain horizontal or vertical.

9. **Ans: (c)** $2s$

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Solution: $F_{\text{rest}} = \left(\frac{GmM}{R^3}\right)r$

$\Rightarrow$ From equation (i) ; $T = 2\pi\sqrt{\frac{R^3}{GM}}$

$\Rightarrow T = 2\pi\sqrt{\frac{R^3}{10R^3}}$

$\Rightarrow T = 2\,\text{sec}$

10. **Ans: (c)** $8\mathrm{P}_1 + \frac{24\,\mathrm{T}}{\mathrm{r}}$

Solution: Using $P_iV_i = P_fV_f$ (for the air bubble) we have

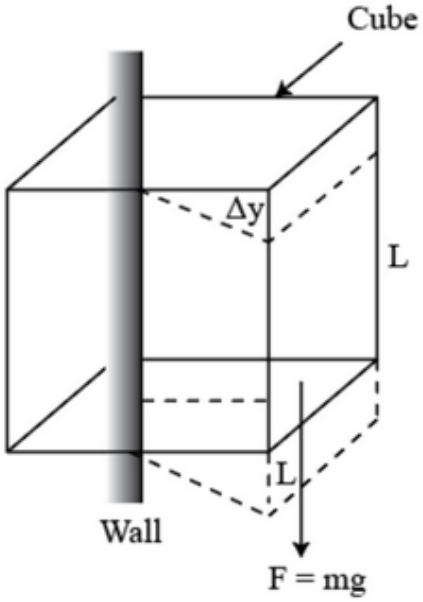
$$\left(\mathrm{P}_1 + \frac{4\,\mathrm{T}}{\mathrm{r}}\right)\left(\frac{4}{3}\pi\mathrm{r}^3\right) = \left(\mathrm{P}_2 + \frac{4\,\mathrm{T}}{\mathrm{r}/2}\right)\left[\frac{4}{3}\pi\left(\frac{\mathrm{r}}{2}\right)^3\right]$$

Solving this, we get

$\mathrm{P}_2 = 8\mathrm{P}_1 + \frac{24\,\mathrm{T}}{\mathrm{r}}$

11. **Ans: (a)** $3.92 \times 10^{-7}\ \mathrm{m}$

Solution:



Given, side of a cube $(1) = 10\ \text{cm} = 0.1\ \text{m}$

Area its each face $(\mathrm{A}) = 1^2 = (0.1)^2 = 0.01\ \mathrm{m}^2$

Load $(m) = 100\ \text{kg}$

Tangential force acting on one face of the cube,

$\mathrm{F} = \mathrm{mg} = 100 \times 9.8 = 980\ \mathrm{N}$

Shear stress acting on this face $= \frac{\mathrm{F}}{\mathrm{A}}$

$= \frac{980}{0.01}\ \mathrm{N/m^2}$

$= 9.8 \times 10^4\ \mathrm{N/m^2}$

Shear modulus of aluminium $(\eta)) = 25\mathrm{GPa}$

$= 25 \times 10^9\ \mathrm{N/m^2}$

Shear modulus $(\eta) = \frac{\text{Shearing stress}}{\text{Shearing strain}}$

or shearing strain $\left(\frac{\Delta y}{L}\right) = \frac{\text{Shearing stress}}{\text{Shear modulus}}$

or $\Delta\mathrm{y} = \frac{\text{Shearing stress}}{\text{Shear modulus}} \times \mathrm{L}$

$= \frac{9.8 \times 10^4}{25 \times 10^9} \times 0.1$

$= 0.392 \times 10^{-5}\ \mathrm{m}$

$= 3.92 \times 10^{-7}\ \mathrm{m}$

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12. **Ans: (b)** $\frac{a}{2}$

Solution: Let the displacement at that instant be y

$$K = \frac{1}{2} m\omega^2 \left(a^2 - y^2\right)$$

$$\frac{3}{4} E = \frac{1}{2} m\omega^2 \left(a^2 - y^2\right)$$

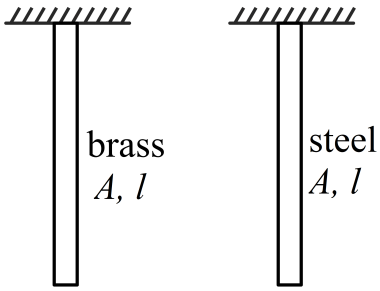
$$\frac{3}{4}\left(\frac{1}{2} m\omega^2 a^2\right) = \frac{1}{2} m\omega^2 \left(a^2 - y^2\right) ,$$

$$y^2 = a^2 - \frac{3}{4} a^2 = \frac{a^2}{4}$$

$$\Rightarrow y = \frac{a}{2}$$

13. **Ans: (a)** 2 : 1

Solution:



$\mathrm{Y} = \frac{\mathrm{W}}{\mathrm{A}} \times \frac{\mathrm{l}}{\Delta \mathrm{l}}$

So, $\Delta \mathrm{l} = \frac{\mathrm{Wl}}{\mathrm{AY}}$

$\Delta \mathrm{e}_1 = \Delta \mathrm{e}_2$

$\frac{\mathrm{w}_1 \mathrm{l}}{\mathrm{AY}_1} = \frac{\mathrm{w}_2 \mathrm{l}}{\mathrm{AY}_2}$

$\Rightarrow \frac{\mathrm{w}_1}{\mathrm{w}_2} = \frac{\mathrm{Y}_1}{\mathrm{Y}_2} = 2$

,

14. **Ans: (d)** 1 s

Solution:

Given, density of bob, $\rho = \frac{4}{3} \times 10^3 \text{ kg m}^{-3}$
Density of water, $\sigma = 10^3 \text{ kg m}^{-3}$

If g' be gravitational acceleration in water then,

$g' = g\left(1 - \frac{\sigma}{\rho}\right) = g\left(1 - \frac{10^3}{\frac{4}{3} \times 10^3}\right) = \frac{g}{4}$

As, $T_{\text{air}} = 2\pi\sqrt{\frac{I}{g}}$,

Similarly, $T_{\text{water}} = 2\pi\sqrt{\frac{I}{g'}}$ $\left[T_{\text{water}} = 2 \text{ s}\right]$ $\left(g' = \frac{g}{4}\right)$

$2 = 2\pi\sqrt{\frac{I}{g}} \cdot 2$

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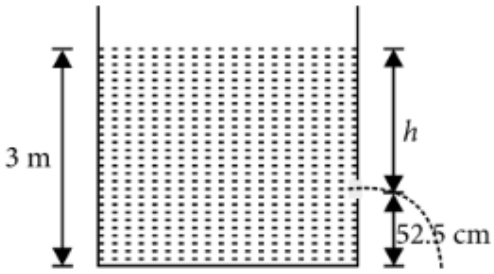
$2 = 2.T_{\text{air}}$

$\Rightarrow \quad T_{\text{air}} = 1\ \text{s}$

;

15. **Ans: (a)** $50\ \text{m}^2\ \text{s}^{-2}$

Solution: Suppose A be the area of cross section of tank, a be the area of hole, v_e be the velocity of efflux, h be the



height of liquid above the hole,
Let v be the speed with which the level decreases in the container. Using equation of continuity, we get $av_e = Av$ or $v = \frac{av_e}{A}$ Using Bernoulli's theorem, we have

$$P_0 + h\rho g + \frac{1}{2}\rho v^2 = P_0 + \frac{1}{2}\rho v_e^2$$

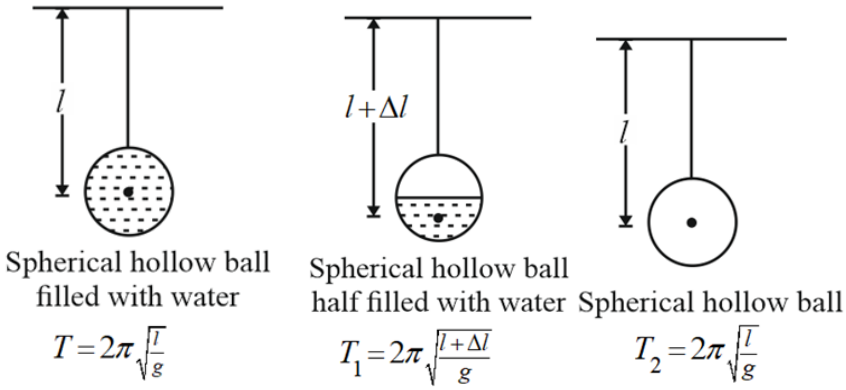
$$h\rho g + \frac{1}{2}\rho\left(\frac{av_e}{A}\right)^2 = \frac{1}{2}\rho v_e^2$$

$$\text{or} \quad v_e^2 = \frac{2hg}{1-\left(a^2/A^2\right)} = \frac{2\times(3-0.525)\times 10}{1-(0.1)^2}$$

$$= 50\ \text{m}^2\ \text{s}^{-2}$$

!

16. **Ans: (a)** First increase and then decrease to the original value



Solution:

and $T_1 > T_2$
The given system is like a simple pendulum whose effective length is between the point of suspension and the center of gravity of the hanging body.
When water slowly flows out, the sphere, the centre of gravity. of system is changing as shown in the figures.
Hence, time period first increases and then decreases to the original value. ^

17. **Ans: (d)** $\frac{p}{3B}$

Solution: $V = \text{Volume of sphere} = \frac{4}{3}\pi r^3$

$\Rightarrow \frac{\Delta V}{V} = 3\frac{\Delta r}{r}$

also, bulk modulus, $B = \left|\frac{-p}{\left(\frac{\Delta V}{V}\right)}\right|$

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$3\frac{\Delta r}{r} = \frac{p}{B}$

$\frac{\Delta r}{r} = \frac{p}{3B}$ /

18. **Ans: (c)** If assertion is true but reason is false

Solution: The excess pressure inside the small drop is large as compared to the large drop because of which smaller drop of liquid resists deforming force better than the large drop. Excess pressure $= \frac{2T}{r}$ where $T =$ surface tension $r =$ radius of liquid drop. Therefore excess pressure is inversely proportional to its radius and hence the surface area. !

19. **Ans: (a)** the motion of the particle is SHM with an amplitude of $\sqrt{A^2 + \frac{B^2}{4}}$

Solution: The displacement of the particle is given by:

$$x = A\sin(-2\omega t) + B\sin^2\omega t$$
$$= -A\sin 2\omega t + \frac{B}{2}(1 - \cos 2\omega t)$$
$$= -\left(A\sin 2\omega t + \frac{B}{2}\cos 2\omega t\right) + \frac{B}{2}$$

This motion represents SHM with an amplitude: $\sqrt{A^2 + \frac{B^2}{4}}$, and mean position $\frac{B}{2}$;

20. **Ans: (b)** $\frac{\rho_2 - \rho}{\rho - \rho_1}$

Solution: $V =$ Volume of solid sphere. Let $V_1 =$ Volume of the part of the sphere immersed in a liquid of density ρ_1 and $V_2 =$ Volume of the part of the sphere immersed in liquid of density ρ_2. According to law of floatation,

$V\rho g = V_1\rho_1 g + V_2\rho_2 g$ ... (i)

and $V = V_1 + V_2$ ... (ii)

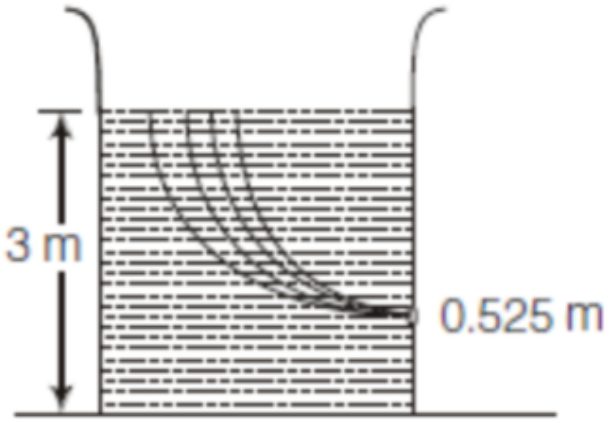
Hence from eqns (i) and (ii),

$$V_1\rho g + V_2\rho g = V_1\rho_1 g + V_2\rho_2 g$$
$$\text{or } V_1(\rho - \rho_1)g = V_2(\rho_2 - \rho)g \quad *$$
$$\text{or } \frac{V_1}{V_2} = \frac{\rho_2 - \rho}{\rho - \rho_1}$$

21. **Ans: 50**

Solution: Let v be velocity of efflux and v' be the velocity of fall of liquid level in tank. Using equation of continuity for tube of flow formed between level of tank and orifice is

$AV' = av$



Applying Bernoulli's theorem, we have $\frac{p}{\rho} + \frac{1}{2}v^2 + gh = \frac{p}{\rho} + \frac{1}{2}v^2 + 0$

$v^2 + \frac{2gh}{1-\left(\frac{a}{A}\right)^2} = \frac{2\times 10\times(3-0.525)}{1-(0.1)^2}$

$= 50\ \text{m}^2/\text{s}^2$

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22. **Ans: 2**

Solution: $\Delta l = \frac{Fl}{AY}$

$\frac{\Delta l}{F/A} = \frac{l}{Y}$

$Y = \frac{4000 \times 10^3}{2 \times 10^{-3}} = 2 \times 10^9 \text{ N/m}^2$

23. **Ans: 5**

Solution: $T = 2\pi\sqrt{\frac{m}{k_1 + k_2}} = 2\pi\sqrt{\frac{10}{360}} = \frac{\pi}{3}\ s$

The maximum velocity is always at equilibrium position since at any other point there will be a restoring force.

Attempting to slow the mass. $\therefore v_{\max} = \frac{\text{impulse}}{\text{mass}} = \frac{50}{10} = 5 \text{ m/s}$

$\Rightarrow \omega = \frac{2\pi}{T} = 6 \text{rad/s}$

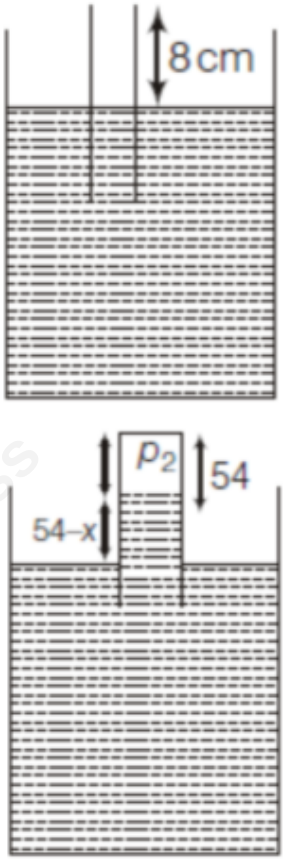
$\Rightarrow A = \text{amplitude} = \frac{v_{\max}}{\omega} = \frac{5}{6} = 0.83 \text{ m}$

24. **Ans: 16**

Solution: In this question, the system is accelerating horizontally, i.e. no component of acceleration in vertical direction. Hence, the pressure in the vertical direciton will remain unaffected.

i.e. $p = p_0 + \rho g h$

Again, we have to use the concept that the pressure in the same level will be same



For air trapped in tube, $p_1 V_1 = p_2 V_2 p_1 = p_{\text{atm}} = \rho g 76$

$\because v_1 = A \cdot 8 (A = \text{area of cross-section})$

$\Rightarrow p_2 = p_{\text{atm}} - \rho g(54 - x) = \rho g(22 + x)$

$\because V_2 = A \cdot x$

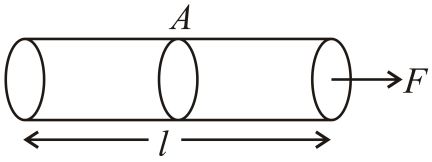
or $\rho g 76 \times 8A = \rho g(22 + x)Ax$

$x^2 + 22x - 76 \times 8 = 0 \Rightarrow x = 16 \text{ cm}$

25. **Ans: 9**

Solution:

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$Y = \frac{Fl}{A\Delta l}$

$\because \ V = Al$ so $l = \frac{V}{A}$

$F = \frac{YA\Delta l}{l} = \frac{YA^2\Delta l}{V}$

$\frac{F_1}{F_2} = \left(\frac{A_1}{A_2}\right)^2 \Rightarrow \frac{F}{F_2} = \left(\frac{A}{3A}\right)^2 = \frac{1}{9}$

$F_2 = 9F_1 = 9F$

26. **Ans: 4**

Solution: Let the velocity of efflux at point 2 is v
Given, $a = 1\ \text{cm}^2, A = 0.5\ \text{m}^2 = 0.5 \times 10^4\ \text{cm}^2$
Applying continuity equation,

$$AV = av$$

$$V = \frac{a}{A}\, v$$

Since, $a \ll A, V \ll v, V^2$ is negligible.
Pressure at (1)

$$p_1 = p_0 + \frac{mg}{A}$$

$$= p_0 + \frac{30 \times 10}{0.5}$$

$$= p_0 + 600$$

$$\text{Pressure at } (2),$$

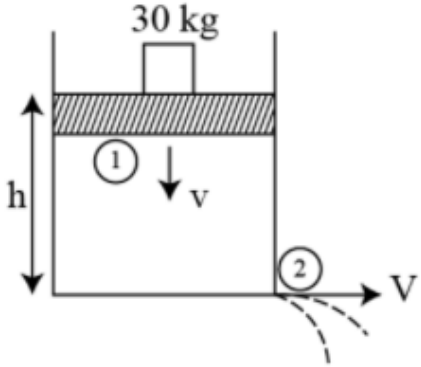
$$p_2 = p_0$$

$$\text{where, } p_0 \text{ is atmosple}$$

where, p_0 is atmospheric pressure.
Applying Bernoulli's equation between (1) and (2)

$p_1 + \rho gh + \frac{1}{2}\,\rho V^2 = \rho_2 + \frac{1}{2}\,\rho v^2$

V^2 is negligible, then



$$p_0 + 600 + 1200 \times 10 \times 0.75$$

$$= p_0 + \frac{1}{2} \times 1200v^2$$

$$600v^2 = 9600$$

$$\Rightarrow v^2 = 16$$

$$\Rightarrow v = 4\ \text{m s}^{-1}$$

27. **Ans: 4**

Solution: For the wire-block system in equilibrium,
Young modulus, $Y = \frac{\text{stress}}{\text{strain}} = \frac{mg/A}{\Delta l/L} = \frac{mg}{A}\,\frac{L}{\Delta l}$

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$\therefore$ weight $mg = \frac{YA}{L}\,\Delta l = k\Delta l$
where $\frac{YA}{L} = k =$ force constant The force equation for the block is, $F = -kx$
$\Rightarrow \quad a = -\frac{k}{m}\,x \Rightarrow$ angular frequency, $\omega = \sqrt{\frac{k}{m}}$
or $\omega = \sqrt{\frac{YA}{Lm}} = \sqrt{\frac{(n)\left(10^9\ \mathrm{N\ m^{-2}}\right)\left(4.9\times10^{-7}\ \mathrm{m^2}\right)}{(1\ \mathrm{m})(0.1\ \mathrm{kg})}} = \left(70\mathrm{rads}^{-1}\right)\sqrt{n}$
But $\omega = 140\mathrm{rads}^{-1}$
$\Rightarrow \quad n = 4$

28. **Ans: 2**

Solution: Stoke's law : $6\pi R\eta v = \frac{4}{3}\,\pi R^3(\rho - \sigma)g \therefore \quad v \propto R^2$. As M is given, $\rho = \frac{M}{\frac{4}{3}\pi R^3}$ But as R^3 increases, M also increases and ρ is a constant. σ also is a constant. $\therefore \quad v$, the terminal velocity $\propto R^2$.

29. **Ans: 25**

Solution:

Force constant K of rubber is given by

$K = \frac{YA}{l} = \frac{5\times10^8\times10^{-6}}{0.1} = 5\times10^3\ \mathrm{N\ m^{-1}}$

Now, from conservation of energy,

elastic potential energy of cord = kinetic energy of particle

i.e. $\frac{1}{2}\,K\left(\Delta l\right)^2 = \frac{1}{2}\,mv^2$

$\Rightarrow v = \sqrt{\frac{K}{m}}\cdot\Delta l$

$= \sqrt{\frac{5\times10^3}{5\times10^{-3}}}(0.125 - 0.1)$

$= 10^3\times0.025 = 25\ \mathrm{m\ s^{-1}}$

30. **Ans: 2**

Solution: The amplitude of oscillations will be the maximum when compression in the spring is maximum. At the time of maximum compression, velocities of both the blocks are equal say v, then using law of conservation of momentum,
$m_1v_0 = (m_1 + m_2)v$
$\Rightarrow 1\times12 = (1+2)v \Rightarrow v = 4\mathrm{cms}^{-1}$
Using law of conservation of energy, we have
$\frac{1}{2}\,m_1v_0^2 = \frac{1}{2}\,kx^2 + \frac{1}{2}\,(m_1 + m_2)v^2$
Putting the value of $v = 4\mathrm{cms}^{-1}$ and solving, we get $x = 2\ \mathrm{cm}.$

31. **Ans: (c)** Rate $= \mathrm{k}[\mathrm{B_2}]$

Solution: For the reaction $2\ \mathrm{A} + \mathrm{B_2} \rightleftharpoons 2\mathrm{AB}$,
Rate $\propto [\mathrm{A}]^{\mathrm{x}}[\mathrm{B_2}]^{\mathrm{y}}$. On substituting the given data, we get
From experimental
$1, 1.6\times10^{-4} \propto [0.50]^{\mathrm{x}}[0.50]^{\mathrm{y}}$ ....(i)
From experimental $2, 3.2\times10^{-4} \propto [0.50]^x[1.00]^y \ldots$ (ii)
From experimental 3, $3.2\times10^{-4} \propto [1.00]^x[1.00]^y \ldots$ (iii)
$1 = \left[\frac{1.00}{0.50}\right]^{\mathrm{x}}$
or $1 = 2^{\mathrm{x}}$ or $2^0 = 2^{\mathrm{x}}$ or $\mathrm{x} = 0$
Now, divide equation (ii) by equation (i) we get,
$2 = \left[\frac{1.00}{0.50}\right]^{\mathrm{y}}$

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$y = 1$

Thus rate equation is: Rate $= k[A]^0[B_2]^1 = k[B_2]$.

32. **Ans: (a)** Statement-1 is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement-1.

Solution:

In the process of reverse osmosis, solvent particles pass through semipermeable membrane towards dilute solution.

In the case of saline water, saline water is more concentrated than that of pure water. Hence water flows from saline solution to pure water.

33. **Ans: (c)** (ii), (iv)

Solution:

$$Fe^{2+} + 2e^- \longrightarrow Fe; E^\circ = -0.44$$
$$Cu^{2+} + 2e^- \longrightarrow Cu; E^\circ = +0.34$$
$$Ag + 1e^- \longrightarrow Ag; E^\circ = +0.80$$

If a metal have negative electrode potential than metal easily displace, other metal, i.e. Fe can displace Cu and Ag.

34. **Ans: (c)** 2.3×10^2 s

Solution: $\text{cis} - z \underset{k_b}{\overset{k_t}{\rightleftharpoons}} \text{trans} - Z$

Initial	a	0
At time t	$a - x$	x
At equilibrium	$a - x_\theta$	x_θ

$K_{aq} = \frac{K_f}{K_b}$

$\therefore K_b = \frac{3 \times 10^{-5}}{10^{-2}} = 3 \times 10^{-3}$

According to relation, $\left(K_f + K_b\right) = \frac{2.303}{t} \log \frac{x_\theta}{x_0 - x}$

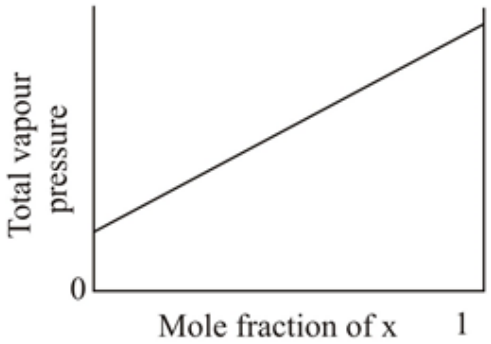
$\left(K_f + K_b\right) = \frac{2.303}{t} \log 2 [\because x = x_e/2]$

$\left(3 \times 10^{-5} + 3 \times 10^{-3}\right) = \frac{0.693}{t}$

$t = \frac{0.693}{3 \times 10^{-3}(1 + 0.01)}$

$= \frac{0.693}{3 \times 10^{-3} \times 101}$

$= 2.3 \times 10^2$ s



35. **Ans: (b)**

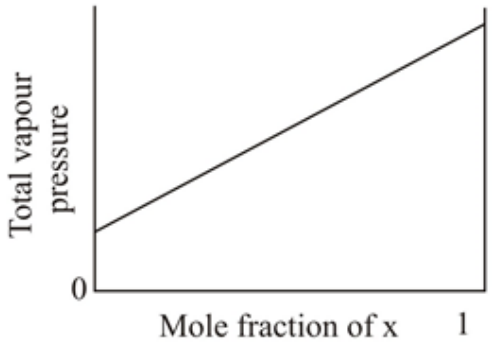
Solution:

By Raoult's law, the vapour pressure of a solvent above a solution is equal to the vapour pressure of the pure solvent at the same temperature scaled by the mole fraction of the solvent present.

$$P_{\text{solution}} = X_{\text{solvent}} P^0_{\text{solvent}}$$

Plot of total vapour pressure as a function of mole fraction of the components of an ideal solution formed by mixing liquids.

Typesetting math: 56%



36. **Ans: (a)** activation energy for the reaction is very high at room temperature

    Solution: The activation energy for the reaction is very high at room temperature hence reaction will not take place.

37. **Ans: (a)** 0.184 g

    Solution: Mass of Au deposited = Number of Faraday passed $\times$ Eq. mass
    $= \frac{0.30 \times 15 \times 60}{96485} \times \frac{197}{3} = 0.184 \text{ g}$

38. **Ans: (d)** no blue colour formation

    Solution: There will be no blue colour formation because in the process of osmosis, solvent molecules migrate across the membrane and not solute molecules.

39. **Ans: (b)** $M^+ + X^- \rightarrow M + X$ is the spontaneous reaction

    Solution: $M^+ + X^- \rightarrow M + X$ is spontaneous because for the cell represented by $M\left|M^+\right|\left|X\right|X$, the value of E° is positive i.e. $(0.44 - 0.33)\text{V} = 0.11 \text{ V}$.

40. **Ans: (c)** 5.88 V

    Solution: Now, the cell reaction is
    $\text{Li(s)} + \frac{1}{2} \text{ F}_2\text{(g)} \rightarrow \text{Li}^+ + \text{F}^-$
    We know that,
    $\text{E}_{\text{cell}} = \text{E}^{\text{o}}_{\text{cell}} = \frac{\text{RT}}{\text{nF}} \ln \frac{\text{Product}}{\text{Reactant}}$
    $= \text{E}^{o}_{\text{cell}} - \frac{2.303\text{RT}}{\text{nF}} \log\left[\text{Li}^+\right]\left[\text{F}^-\right]$
    $= 5.92 - \frac{2.303 \times 8.314 \times 298}{1 \times 96500} \log(2 \times 2)$
    $= 5.92 - \frac{0.059}{1} \times 2 \log 2$
    $= 5.92 - 0.035 = 5.887 \text{ V}$

41. **Ans: (b)** 2 and 3

    Solution:

    In presence of acid, hydrolysis of ethyl acetate is a pseudo-unimolecular reaction

    Pseudo first order rate constant

    $\text{k'} = \text{k}\left[\text{H}^+\right]$

    $\text{rate } \left(\text{r}\right) = \text{k}\left[\text{H}^+\right][\text{ester}]$

    If the acid concentration $\left[\text{H}^+\right]$ is doubled at constant ester concentration, then

    1.The second order rate constant, k is remains same

    2.The pseudo first order rate constant, K' is double.
    3. The rate of the reaction is doubled

Typesetting math: 56% mol^{-1}

Solution: $T_b^{\circ}=353.23\ \mathrm{K}, W_B=1.8\ \mathrm{g}, W_A=90\ \mathrm{g}, T_b=354.11\ \mathrm{K}, K_b=2.53\ \mathrm{kgmol}^{-1}$
$\Delta T_b=T_b-T_b^{\circ}=354.11-353.23=0.88\ \mathrm{K}\ M_B=\frac{W_B\times K_b\times 1000}{\Delta T_b\times W_A}=\frac{1.8\times 2.53\times 1000}{0.88\times 90}$
$=57.5\approx 58\ \mathrm{g\ mol}^{-1}$

43. **Ans: (a)** I only

Solution:

In any mechanism of a reaction, the slowest step of reaction is rate determining step.

As per mechanism (I), $\mathrm{rate}=\mathrm{k}\ [\mathrm{NO_2}]\ [\mathrm{O_3}]$ -----(1)
As per mechanism (II), $\mathrm{rate}=\mathrm{k}\ [\mathrm{NO_2}]\ [\mathrm{O}]$ ----------(2)

The intermediate $[\mathrm{O}]$ in the above reaction can be eliminated using the following equilibrium reaction

$$\mathrm{O_3(g)}\ \rightleftharpoons \mathrm{O_2\ (g)}\ +\ \mathrm{O}$$

Equilibrium constant
$\mathrm{K_{eq}}=\frac{[\mathrm{O_2}][\mathrm{O}]}{[\mathrm{O_3}]}$
or $[\mathrm{O}]=\mathrm{K_{eq}}\ \frac{[\mathrm{O_3}]}{[\mathrm{O_2}]}$

Put in equation in (2)

$$\mathrm{rate}=\mathrm{k\ K_{eq}}\ [\mathrm{NO_2}]\ [\mathrm{O_3}]\ [\mathrm{O_2}]^{-1}$$

44. **Ans: (c)** $\mathrm{SnCl_4}$

Solution: $\therefore w=\frac{Eit}{F}=\frac{\frac{A}{n}\times i\times t}{F}$
$\Rightarrow \quad n=\frac{A\times i\times t}{w\times F}=\frac{118.7\times 2\times(5\times 3600)}{11.1\times 96500}=3.989$
$A=$ atomic weight of $\mathrm{Sn}=118.7\ \mathrm{g\ mol}^{-1}$
$i=$ current $=\frac{200\ \mathrm{W}}{100\ \mathrm{V}}=2\ \mathrm{A}$
$t=$ time $=5\mathrm{hrs}(5\times 3600\ \mathrm{s})$
$E=\frac{A}{n}=$ equavalent weight of Sn^{n+}
$n=$ valency of Sn in the salt
$w=$ mass of Sn deposited $=11.1\ \mathrm{g}$
Here, SnO and $\mathrm{SnO_2}$ are non-electrolytes (aqueous).
So, valency of Sn will be 4 in $\mathrm{SnCl_4}\left(\mathrm{Sn}^{4+}+4\mathrm{Cl}^{-}\right)$.

45. **Ans: (c)** 2.0

Solution:

$$\begin{aligned}&(\mathrm{c}): \text{Given}: T_1=27^{\circ}\mathrm{C}=27+273=300\ \mathrm{K}\\&T_2=37^{\circ}\mathrm{C}=37+273=310\ \mathrm{K}\\&\text{and } E_a=53.6\ \mathrm{kJ\ mol}^{-1}\\&\log\left(\frac{k_2}{k_1}\right)=\frac{E_a}{2.303R}\left(\frac{T_2-T_1}{T_1T_2}\right)\\&\log\left(\frac{k_{310\ \mathrm{K}}}{k_{300\ \mathrm{K}}}\right)=\frac{53.6\times 10^3}{2.303\times 8.314}\left(\frac{310-300}{300\times 310}\right)\\&=0.3010\\&\therefore \frac{k_{310\ \mathrm{K}}}{k_{300\ \mathrm{K}}}=2\end{aligned}$$

46. **Ans: (c)** 12

Typesetting math: 56%

$\Lambda_{\mathrm{m}}=1000 \times \frac{\mathrm{K}}{\mathrm{M}}$
$=1000 \times \frac{2 \times 10^{-5}}{0.001}=20 \mathrm{~S} \mathrm{~cm}^2 \mathrm{~mol}^{-1}$

$\Rightarrow \alpha=\frac{\Lambda_{\mathrm{m}}}{\Lambda_{\mathrm{m}}^{\infty}}=\frac{20}{190}=\left(\frac{2}{19}\right)$

$=12.3 \times 10^{-6}$

47. **Ans: (b)** If both assertion and reason are true but reason is not the correct explanation of assertion.

Solution: Non-ideal solutions with positive deviation i.e., having more vapour pressure than expected, boil at lower temperature while those with negative deviation boil at higher temperature than those of the components.

48. **Ans: (b)** $\mathrm{A}, \mathrm{B}, \mathrm{C}$

Solution: The more the reduction potential, the more is the deposition of metals at the cathode.
Cation having E° value less than -0.83 V (reduction potential of $\mathrm{H_2O}$) will not deposit from aqueous solution.
$\mathrm{E}_{\mathrm{A}^{3+} / \mathrm{A}}{ }^{\circ}=1.50 \mathrm{~V}$ will deposit at the cathode
$\mathrm{E}_{\mathrm{B}^{2+} / \mathrm{B}}{ }^{\circ}=0.3 \mathrm{~V}$ will deposit at the cathode
$\mathrm{E}_{\mathrm{C}^{3+} / \mathrm{C}}=-0.74 \mathrm{~V}$ will deposit at the cathode
$\mathrm{E}_{\mathrm{D}^{2+} / \mathrm{D}}^{\circ}=-2.37 \mathrm{~V}$ will not deposit at the cathode.

49. **Ans: (c)** Higher the value of $\mathrm{K_H}$ at a given pressure, higher is the solubility of the gas in the liquids

Solution: $\mathrm{p}=\mathrm{K_H} \times \mathrm{x}$
Where $K_H=$ Henry law constant
Henry's constant increases with increase in temperature while it decreases with increase in solubility. Hence, value of K_H increases as solubility of gas increases is a wrong statement.

50. **Ans: (b)** Its examples are- Rusting of iron, tarnishing of silver

Solution:

Corrosion is the slow process of the destruction of materials by a chemical reaction to the environment. The examples of corrosion are rusting of iron, tarnishing of silver.

51. **Ans: 3**

Solution: $2 \mathrm{~N_2O_5} \rightarrow 4 \mathrm{NO_2}+\mathrm{O_2}$
From the unit of rate constant it is clear that the reaction follow first order kinetics.
Hence by rate law equation, $\mathrm{r}=k[\mathrm{~N_2O_5}]$
where $r=1.02 \times 10^{-4}, k=3.4 \times 10^{-5}$
$1.02 \times 10^{-4}=3.4 \times 10^{-5}[\mathrm{~N_2O_5}]$
$[\mathrm{N_2O_5}]=3 \mathrm{M}$

52. **Ans: 1**

Solution:

$\Delta \mathrm{T_f}=\mathrm{iK_fm}$
$0.0557=\mathrm{i} \times 1.86 \times 0.01$
$\mathrm{i}=3$, means complex will give 3-ions

Hence, formula should be $\left[\mathrm{Co(NH_3)_5Br}\right] \mathrm{Br_2}$

53. **Ans: 1**

Solution: Writing the equation for pentane-oxygen fuel cell at respective electrodes and overall reaction, we get
At Anode:
$\mathrm{C_5H_{12}}+10 \mathrm{H_2O} \rightarrow 5 \mathrm{CO_2}+32 \mathrm{H}^{+}+32 \mathrm{e}^{-}$
(pentane)
At Cathode:
$8 \mathrm{O_2}+32 \mathrm{H}^{+}+32 \mathrm{e}^{-} \rightarrow 16 \mathrm{H_2O}$
$8 \mathrm{O_2} \rightarrow 5 \mathrm{CO_2}+6 \mathrm{H_2O}$

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Calculation of ΔG° for the above reaction

$\Delta G^{\circ}=[5 \times(-394.4)+6 \times(-237.2)]$

$-[-8.2]$

$=-1972.0-1423.2+8.2=-3387.0 \mathrm{~kJ}$

$=-3387000$ Joules.

From the equation we find $\mathrm{n}=32$

Using the relation, $\Delta G^{\circ}=-nFE_{cell}^{\circ}$ and substituting various values, we get

$-3387000=-32 \times 96500 \times E_{\text{cell}}^{\circ}(\mathrm{F}=96500 \mathrm{C})$

or $E_{\text{cell}}^{\circ}=\frac{3387000}{32 \times 96500}$

$=\frac{3387000}{3088000}$ or $\frac{3387}{3088} \mathrm{~V}=1.0968 \mathrm{~V}$

54. **Ans: 108**

Solution:

Given $\mathrm{A}=\mathrm{t}_{1/2}=54 \mathrm{~min}$

$\mathrm{B}=\mathrm{t}_{1/2}=18 \mathrm{~min}$

$\mathrm{t}=0$ 'x' $\mathrm{M} \quad \mathrm{t}=0$ 'x' M

$\Rightarrow$ To calculate : $\left[\mathrm{A}_{\mathrm{t}}\right]=16 \times\left[\mathrm{B}_{\mathrm{t}}\right]$ time $=?$

$\Rightarrow$ For I order kinetic : $\left[\mathrm{A}_{\mathrm{t}}\right]=\frac{\mathrm{A}_0}{(2)^{\mathrm{n}}}$

$\mathrm{n} \rightarrow$ no. of Half lives

$\Rightarrow$ Now from the relation (1)

$\Rightarrow \frac{\mathrm{x}}{(2)^{\mathrm{n}_1}}=\frac{\mathrm{x}}{(2)^{\mathrm{n}_2}} \times 16 \Rightarrow(2)^{\mathrm{n}_2}=(2)^{\mathrm{n}_1} \times(2)^4$

$\Rightarrow \mathrm{n}_2=\mathrm{n}_1+4 \Rightarrow \frac{\mathrm{t}}{\left(\mathrm{t}_{1/2}\right)_2}=\frac{\mathrm{t}}{\left(\mathrm{t}_{1/2}\right)_1}+4$

$\Rightarrow \mathrm{t}\left(\frac{1}{18}-\frac{1}{54}\right)=4 \Rightarrow \mathrm{t}=\frac{4 \times 18 \times 54}{36}$

$\Rightarrow \mathrm{t}=108 \mathrm{~min}$

55. **Ans: 6**

Solution:

$\Delta \mathrm{G}^{\circ}=-\mathrm{nF} \mathrm{E}_{\text{cell}}^{\mathrm{o}}$

$17.37 \times 10^3=-3 \times 96500 \times \mathrm{E}_{\text{cell}}^{\mathrm{o}}$

$\mathrm{E}_{\text{cell}}=-0.06 \mathrm{~V}$

56. **Ans: 3**

Solution: $\left[\mathrm{CH}_3\right]=\frac{10^{-5}}{10} \mathrm{~mol} \mathrm{~L}^{-1}$

$=10^{-6} \mathrm{~mol} \mathrm{~L}^{-1}$

$\left[\mathrm{C}_2 \mathrm{H}_6\right]=\frac{100}{10}=10 \mathrm{~mol} \mathrm{~L}^{-1}=10 \mathrm{M}$

$K_{\text{eq}}=\frac{K_f}{K_b}=\frac{\left[\mathrm{CH}_3\right]^2}{\left[\mathrm{C}_2 \mathrm{H}_6\right]}$

$\frac{\left[10^{-6}\right]^2}{10}=\frac{3.14 \times 10^2 \mathrm{~s}^{-1}}{K_b}$

$K_b=3.14 \times 10^{15} \mathrm{~L} \mathrm{~mol}^{-1} \mathrm{~s}^{-1}$

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57. **Ans: 106**

Solution: Molar mass of solute $= \frac{\mathrm{K_f \times w \times 1000}}{\Delta \mathrm{T_f \times W}}$
Given : $\mathrm{K_f} = 1.86, \mathrm{w} = 1.25\ \mathrm{g},\ \mathrm{W} = 20\ \mathrm{g},$
$\Delta \mathrm{T_f} = 273 - 271.9 = 1.1\ \mathrm{K}$
Therefore, molar mass of solute
$= \frac{1.86 \times 1.25 \times 1000}{1.1 \times 20} = 105.7$

58. **Ans: 3**

Solution: $\mathrm{E_{metal}} = (\text{ Wt of metal } \times 96500) / \text{ No of coulombs}$
$= \frac{(22.2 \times 96500)}{(2 \times 5 \times 60 \times 60)} \approx 59.5$
Oxidation no. of metal $= \frac{177}{59.5} = +3$

59. **Ans: 16**

Solution:

Suppose $R1 =$ initial rate and $T1 =$ initial temperature

When temperature is increased from $T1$ to $\mathrm{T1} + 10^{\circ}\mathrm{C}$ the rate doubles; rate $= 2R1$

When temperature is increased from $\mathrm{T1} + 10^{\circ}\mathrm{C}$to $\mathrm{T1} + 20^{\circ}\mathrm{C}$the rate doubles; rate $= 2(\mathrm{2R1}) = \mathrm{4R1}$

When temperature is increased from $\mathrm{T1} + 20^{\circ}\mathrm{C}$ to $\mathrm{T1} + 30^{\circ}\mathrm{C}$the rate doubles;

rate $= 2(4\ R1) = 8\ R1$

When temperature is increased from $\mathrm{T1} + 30^{\circ}\mathrm{C}$ to $\mathrm{T1} + 40^{\circ}\mathrm{C}$ the rate doubles;

rate$= 2(8\ R1) = 16\ R\ 1$

When the temperature is raised by $40^{\circ}\mathrm{C}$ the speed of the reaction increases to about 16 times.

60. **Ans: 125**

Solution:

Let initial moles of HA is a

$\mathrm{2HA} \rightarrow \mathrm{H_2\ A_2}$
$\mathrm{0.5a} \qquad \mathrm{0.25a}$

$\mathrm{HA} \rightleftharpoons \mathrm{H^+ + \ A^-}$
$\mathrm{0.5a\ \ 0.5a\ \ 0.5a}$

Now, $\mathrm{i} = \frac{\text{final moles}}{\text{initial moles}} = \frac{\mathrm{0.25a + 0.5a + 0.5a}}{\mathrm{0.5a + 0.5a}}$

$= 1.25 = 125 \times 10^{-2}$

61. **Ans: (d)** None of these

Solution:

Given, $S(K,\ 0)$ and $S'(-K,\ 0)$,

Now, $SS' = |2K|$ and $PS + PS' = 4$ (given)

Now for the ellipse we know that,

$PS + PS' > SS'$

$\Rightarrow 4 > |2K|$

Typesetting math: 56% $(-2,\ 2)$.

If P, S & S' are collinear, then

$PS + PS' = SS'$

$\Rightarrow K = \pm 2.$

If $PS + PS' < SS'$, then no locus exists.

$\Rightarrow K \in (-\infty,\ -2) \cup (2,\ \infty).$

62. **Ans: (c)** $(4a, 0)$

Solution:

The given parabola is $y^2 = 4ax.$

Let $A = \left(at_1^2, 2at_1\right)$ and

$B = \left(at_2^2, 2at_2\right)$ are two points on it.

Now, slope of $OA = m_{OA} = \frac{2at_1 - 0}{at_1^2 - 0} = \frac{2}{t_1}$

and slope of $OB = m_{OB} = \frac{2at_2 - 0}{at_2^2 - 0} = \frac{2}{t_2}$

$\because OA \perp OB$

$\therefore m_{OA} \times m_{OB} = -1,$

Thus, $t_1 t_2 = -4$

Using $y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1),$

to get the equation of line AB as

$y - 2at_1 = \frac{2a\left(t_2 - t_1\right)}{a\left(t_2^2 - t_1^2\right)}\left(x - at_1^2\right)$

$\Rightarrow y - 2at_1 = \frac{2\left(t_2 - t_1\right)}{\left(t_2 - t_1\right)\left(t_2 + t_1\right)}\left(x - at_1^2\right)$

$\Rightarrow y - 2at_1 = \frac{2}{\left(t_2 + t_1\right)}\left(x - at_1^2\right)$

$\Rightarrow y(t_2 + t_1) - 2at_1 t_2 - 2at_1^2 = 2x - 2at_1^2$

Putting the value of $t_1 t_2 = -4,$

we get

$y(t_2 + t_1) + 8a = 2x$

$\Rightarrow y(t_1 + t_2) = 2(x - 4a),$ which is a family of lines $L_1 + \lambda L_2 = 0,$

where $\lambda = t_1 + t_2.$

Hence, the two lines are

$L_1 \equiv x - 4a = 0,\ L_2 \equiv y = 0$

The point of intersection of these two lines is $(4a, 0)$. Hence, AB

always passes through a fixed point $(4a, 0)$.

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63. **Ans: (b)** straight line

Solution: $x + a(2y + 1) = 0$
$x + b(3y + 1) = 0$
$x + a(4y + 1) = 0$
$$\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4a & a \end{vmatrix} = 0$$
$R_2 \rightarrow R_2 - R_1$
$R_3 \rightarrow R_3 - R_1$

or
Hence, locus of is or

64. **Ans: (c)**

Solution:

Given circles and

Thus, centers of two circles are and radii are respectively.

Since, center of circles are on the opposite side of line .

Since, circles should not intersect given line.

Hence,

or

and or

Thus,

65. **Ans: (a)** at most one

Solution: Let two points and lie on the circumference of a circle with centre where
Point equidistant from & will lie on the perpendicular bisector of this line segment.

where
Now it must pass through

There is no such line possible, hence atmost one point lies on the circle.

66. **Ans: (d)**

Solution:

Let be inclined to transverse axis at an angle such that lies on the hyperbola. Therefore,

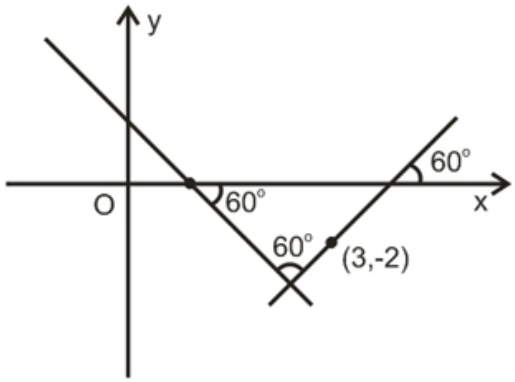
Replacing we get,

(where,)

67. **Ans: (b)**

Solution:

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Slope of line=

Let slope of second line=m

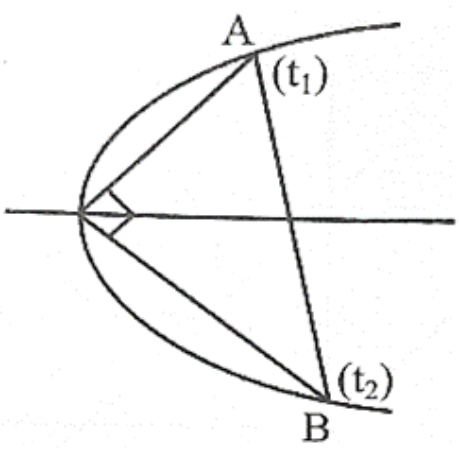
For angle between two lines ,

But line intersects Xaxis, therefore

Slope of req. line
Eq. is
i.e.

68. **Ans: (a)** units

Solution:



Since the chord subtends a right angle at the origin,

Now, the point lies on the parabola,

i.e.

So,
Hence, the length of the chord units

69. **Ans: (b)** Minimum value of

Solution:

Given , and

So, the coordinates of foci and are and respectively.

Let, be a variable point on the ellipse.
Then,
Since the value of lies between and
So, the maximum value of area of is sq. units
Also,

Typesetting math: 56%

70. **Ans: (b)**

Solution: Let, the lines are and
which intersects at the point
&

(given,)
(Hyperbola)
Eccentricity

71. **Ans: (b)**

Solution:

If is the midpoint of the segment between the axes, them point and .

The equation of the line is,

or,

The line perpendicular to it is of form,

Since it passes through ,

Line is,

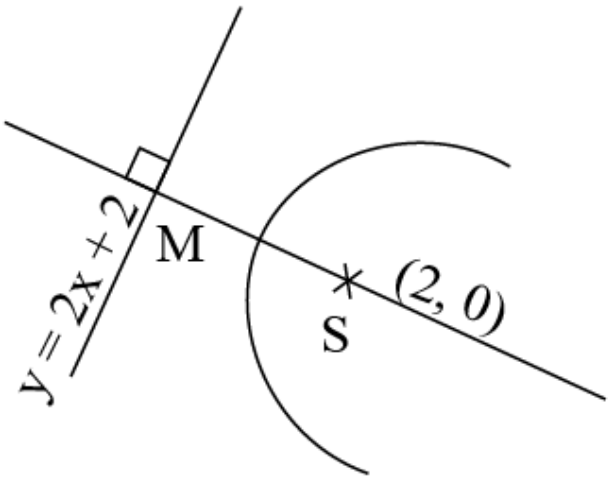
Solving and we get .

72. **Ans: (a)** \frac{325}{36}

Solution: \begin{array}{l} 4 x^{2}+9 y^{2}-8 x-36 y+15=0 \\ 4\left(x^{2}-2 x\right)+9\left(y^{2}-4 y\right)=-15 \\ 4\left(x^{2}-2 x+1\right)+9\left(y^{2}-4 y+4=-15+4+36\right. \\ 4(x-1)^{2}+9(y-2)^{2}=25 \\ \frac{(x-1)^{2}}{\left(\frac{5}{2}\right)^{2}}+\frac{(y-2)^{2}}{\left(\frac{5}{3}\right)^{2}}=1 \ldots \ldots \ldots \ldots \ldots \ldots \ldots . .(1) \\ x^{2}-2 x+y^{2}-4 y+5 \\ (x-1)^{2}+(y-2)^{2} \end{array} \min of \left((x-1)^{2}+(y-2)^{2}\right)=\frac{25}{9} \max of \left((x-1)^{2}+(y-2)^{2}\right)=\frac{25}{4} =\frac{25}{9}+\frac{25}{4}=\frac{325}{36}

73. **Ans: (a)** \frac{\sqrt{5}}{12}

Solution:



Length of \mathrm{LR}=2(\mathrm{SM})
=2 \frac{|2(2)+2-0|}{\sqrt{2^{2}+1}}=\frac{12}{\sqrt{5}}
\Rightarrow \frac{1}{\mathrm{~K}}=\frac{12}{\sqrt{5}} \Rightarrow \mathrm{K}=\frac{\sqrt{5}}{12}

74. **Ans: (a)** 1

Solution:

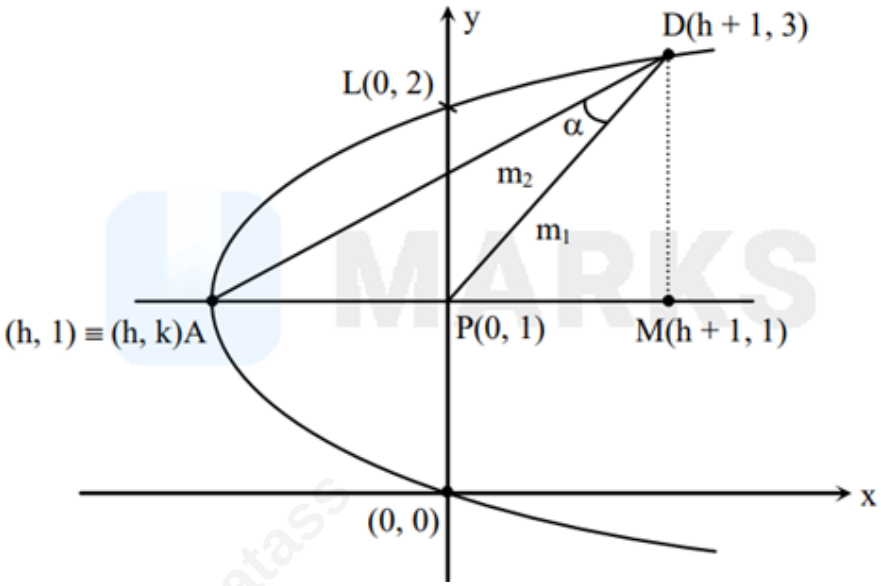
Let and are the vertice of the and D, E and F are the midpoints,As we know that

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75. **Ans: (c)** \mathrm{x}^{2}+\mathrm{y}^{2}=4 \mathrm{a}^{2}

Solution: We know that centroid divides the median in the ratio 2: 1.
Radius of the circle =\frac{2}{3} \times length of median =\frac{2}{3} \times 3 a=2 a
Centre of the (given) circle is \mathrm{C}(0,0).
Therefore the equation of the circle (x-0)^{2}+(y-0)^{2}=(2 a)^{2} \Rightarrow x^{2}+y^{2}=4 a^{2}

76. **Ans: (b)** \tan ^{-1} \frac{2}{19}

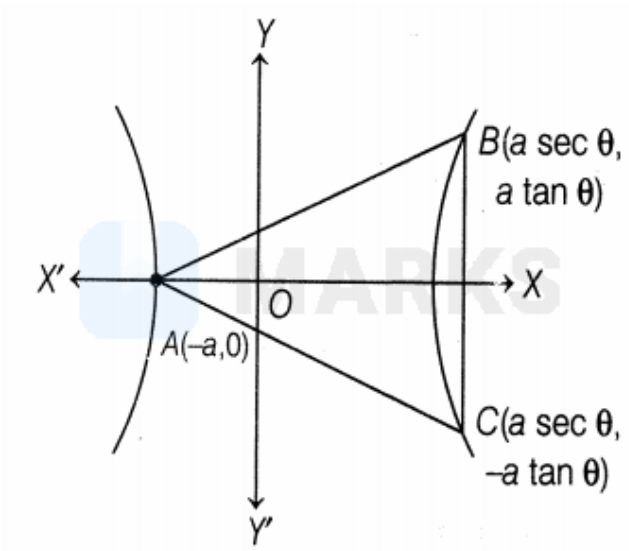


Solution:
Curve, S:(y-k)^{2}=4(x-h) LLR =4 ; Clearly \mathrm{k}=1 ; \Rightarrow \mathrm{A}(\mathrm{h}, 1) \& ' \mathrm{M}^{\prime} is focus (\mathrm{h}+1,1) So D (\mathrm{h}+1,3) \mathrm{S}_{(0,0)}=0 \Rightarrow \mathrm{k}^{2}=-4 \mathrm{~h} \Rightarrow \mathrm{h}=\frac{-1}{4} \quad \Rightarrow \mathrm{D}\left(\frac{3}{4}, 3\right) Now ; \tan \alpha=\left|\frac{m_{1}-m_{2}}{1+m_{1} m_{2}}\right|=\left|\frac{\frac{8}{3}-2}{1+\frac{8}{3} \times 2}\right|=\frac{2}{19} where, m_{1}=\frac{3-1}{\frac{3}{4}-0}=\frac{2}{\frac{3}{4}}=\frac{8}{3} \mathrm{m}_{2}=\frac{3-1}{1}=2

77. **Ans: (b)**

Solution:

We have rectangular hyperbola



Given is an equilateral triangle.

Side

But side of triangle is .

Hence,

Typesetting math: 56%

Solution:

We have,

Hence, slope is . Therefore,

The coordinate of

Let be the image of w.r.t. to the line , then

Image of w.r.t. the line is .

79. **Ans: (a)** .

Solution:

Since, length does not vary with change of axes or shifting of origin.
So, for simplicity,

consider
Now, focus is So, focal chord at a distance of units from the vertex is latus rectum which has length .
So the length of required focal chord will be .

80. **Ans: (c)**

Solution: Let and

Right angle at

81. **Ans: 6**

Solution: Any point on the hyperbola is
Now, the image of by the line mirror is

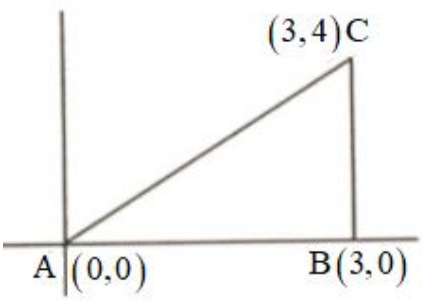
and
Now eliminating from above two equations we get,

82. **Ans: 1**

Solution: is
is
is the only common point

83. **Ans: 7**

Solution:



Also

u/wrapthatass

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Length of the latus rectum, units

84. **Ans: 6**

Solution:

Reflection of about the line is which lies on side
Equation of is {since, line is passing through and }
is the point of intersection of and

coordinates of is
coordinates of is { mid point of and is }

85. **Ans: 72**

Solution: Let is then is
mid point of is

{ where }

86. **Ans: 3**

Solution: We know, area of a triangle
\Delta=\frac{1}{2} \times base \times altitude
\because \frac{\left(\frac{1}{p_{1}}\right)^{2}+\left(\frac{1}{p_{3}}\right)^{2}}{\left(\frac{1}{p_{2}}\right)^{2}}=\frac{\left(\frac{2 \Delta}{p_{1}}\right)^{2}+\left(\frac{2 \Delta}{p_{3}}\right)^{2}}{\left(\frac{2 \Delta}{p_{2}}\right)^{2}}=\frac{(B C)^{2}+(A B)^{2}}{(A C)^{2}} {where \Delta represents the area of triangle A B C\}
=\frac{(5-3)^{2}+(2-(-1))^{2}+(-1-5)^{2}+(1-2)^{2}}{(-1-3)^{2}+(1-(-1))^{2}}
=\frac{4+9+36+1}{16+4}=\frac{50}{20}=2.5

87. **Ans: 9**

Solution:

The equation of the common chord is
which passes through
i.e.

88. **Ans: 6**

Solution:

89. **Ans: 4**

Solution:

Let and be two rods of length and sliding along axis and axis respectively.

Let the equation of circle through and

intercept and circle

i.e.

intercept and circle

i.e.

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Locus of centre is

90. **Ans: 4851**

Solution: \# of points on x-axis: 0
\# of points on the line y=1 : Still 0.
Because both the points (0,1) and (1,1) are ON the triangle and NOT INSIDE the triangle
\# of points on the line y=2 is 1 : The point being (1,2)
\# of points on the line y=3 is 2 : The points being (1,3),(2,3)
\# of points on the line y=4 is 3 : The points being (1,4),(2,4),(3,4)
\# of points on the line y=99 is 98. The points being (1,99),(2,99),(3,99) \ldots(98,99)
Hence, total answer: 0+0+1+2+3+\ldots+98
=98 * 99 / 2

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